

Apoorva Nandigama

Electrical & Wireless Engineer | Hardware Validation & Python Automation

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F-1 OPT authorized through Feb 2027 (STEM OPT eligible)

Profile Summary

- IEEE NAECON 2025 published MS Electrical Engineering graduate (University of Cincinnati, GPA 3.37/4.0) with hands-on experience in electrical circuit validation, bench instrumentation, PLC/HMI programming, and Python test automation.
- Currently performing component-level circuit validation and PLC programming in an active automotive manufacturing environment at Mitsubishi Electric Automotive America.
- Strongest in roles where I test, debug, document, and support engineering and cross-functional teams — with additional depth in wireless systems (5G NR, OFDM, MIMO, Rayleigh fading) from published research (Siemens industry collaboration, IEEE NAECON 2025)

Technical Skills

Bench & Hardware: Oscilloscope, signal generator, multimeter, power supply; RLC filters, MOSFET motor drive, crowbar overvoltage protection, optocoupler isolation; Mitsubishi Q-Series PLC (GX Works2); iQ-R servo commissioning

Embedded Systems: Arduino UNO (ATmega328P), embedded C firmware; GSM SIM900, HC-05 Bluetooth; capacitive sensors; systematic fault isolation

Programming: Python (NumPy, SciPy, pandas, Matplotlib), MATLAB, LabVIEW, embedded C, Git, Linux CLI

ML & AI: PyTorch (DQN reinforcement learning), TensorFlow, scikit-learn, OpenCV, CUDA 12.5, Open3D

Wireless & RF: 5G NR (PHY/MAC/RLC/PDCP/RRC), OFDM, MIMO, HARQ, Rayleigh fading, Doppler analysis, RF link budget, CDMA/SIC, BLE, Wi-Fi 802.11ac/ax

Test & Simulation: Automated test execution, data pipelines, statistical analysis (P50–P99, CDF), Falstad simulation, KiCad, SolidWorks, Monte Carlo simulation, DFT/FFT, structured engineering reports

Work Experience

Training & Development Engineering Intern

Mitsubishi Electric Automotive America

Feb 2026 – Present

Mason, OH

- Performing component-level circuit validation in an active automotive manufacturing environment — building and bench-testing RLC filters, MOSFET motor drive, crowbar overvoltage protection, and optocoupler isolation circuits.
- Measuring electrical behavior using oscilloscopes, signal generators, and multimeters; documenting test conditions, observed results, and structured root cause analysis.
- Programmed Mitsubishi Q-Series PLC (GX Works2) for production automation; independently expanded to iQ-R platform (CC-Link IE Field) for servo commissioning.

Research Assistant — Electrical Engineering

University of Cincinnati

Feb 2024 – Dec 2024

Cincinnati, OH

- Built Python automated test pipeline in Siemens collaboration — NumPy, SciPy, Open3D on Tesla V100; 2,400+ configurations, pass/fail validation, metric extraction, statistical analysis, and structured engineering reports.
- Research achieved 90% bandwidth reduction with F-score 88.774%; published IEEE NAECON 2025 (DOI: 10.1109/NAECON65708.2025.11235332).

IoT Hardware Validation Intern

Pentagon Rugged Systems

Jul 2022 – Nov 2022

India

- Designed and validated an embedded IoT hardware system integrating Arduino UNO, GSM SIM900, HC-05 Bluetooth, capacitive sensors, and LCD — 200+ lines embedded C firmware achieving ~95% detection accuracy and sub-3-second alert latency.
- Diagnosed root cause firmware failure through systematic fault isolation (SIM900 PDU mode — AT+CMGF=1 fix).

Test Automation Intern

VI Solutions

May 2021 – Jun 2021

India

- Built LabVIEW automated test framework for embedded circuit validation: signal acquisition, pass/fail logic, and automated test report generation.

RF Link Budget Intern

NPHSAT Systems

May 2021 – Jun 2021

India

- Computed RF link budgets for satellite telemetry downlink channels — path loss, noise figure, SNR margins, and receiver sensitivity across atmospheric and propagation scenarios.

Projects & Research

IEEE NAECON 2025 Publication — Siemens Collaboration

- Published Efficient Point Cloud Streaming via Curvature and Edge-Aware Sampling — IEEE NAECON 2025 (DOI: 10.1109/NAECON65708.2025.11235332).
- Achieved 90% bandwidth reduction and F-score 88.774% across 2,400 test configurations and 8 ShapeNet categories.
- Additional publications: ANN Weather Monitoring (IDCIoT 2023, DOI: 10.1109/IDCIoT56793.2023.10053534); Smart Weapon Detection (IGI Global).

Tools: Python · NumPy · SciPy · Open3D · CUDA 12.5 · Tesla V100

DQN Adaptive Modulation — AI for Next-Generation Wireless Networks

- Developed a PyTorch DQN agent for adaptive modulation and coding scheme selection — agent observes real-time channel state (SNR, bandwidth, power) and autonomously selects QPSK, 16-QAM, or 64-QAM, replacing static threshold-based link adaptation. Reward converged above 1.10 after 5,000 training episodes.

Tools: PyTorch · Python · Reinforcement Learning

Rayleigh Fading Channel Simulation — 5G NR Performance

- Built a multipath channel simulation in MATLAB/Python — BER vs Eb/N0, SNR degradation, and ICI analysis under varying mobility and Doppler conditions.

Tools: MATLAB · Python

Monte Carlo Radar Detection Framework

- Python statistical framework with 1,000+ Monte Carlo runs, automated P50–P99 reporting, and CDF visualization for radar detection performance analysis.

Tools: Python · NumPy · Matplotlib

Stair-Climbing Wheelchair Robot — Electromechanical Design

- Designed and assembled an electromechanical prototype with Rocker-Bogie suspension, servo actuators, and HC-05 Bluetooth control — 200+ lines embedded C/Arduino firmware.

Tools: SolidWorks · Arduino · Embedded C · HC-05 Bluetooth

Education

M.S. in Electrical Engineering

Aug 2023 – Dec 2025

University of Cincinnati, OH · GPA: 3.37/4.0

Specialization: Wireless Communications · Coursework: Wireless Communications, DSP, ML for Engineers, Probability & Random Processes

B.Tech. in Electronics & Communication Engineering

Jul 2019 – May 2023

KLEF University, India · GPA: 9.08/10.0

Specialization: Wireless Communication

Certifications

- NI CLAD — LabVIEW Associate Developer Credential | NI CoE, KLEF University | 2020 | 40 Hours
- Hybrid Electric Vehicle Technology — Basics | Toyota Kirloskar Motor | May 2025
- Generative AI Fundamentals | Databricks | March 2025
- HCIA-AI Certification | Huawei | 2022
- Wireless Communications for Everybody | Yonsei University (Coursera) | 2020